

Assignment: 01

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SQA Batch: 21

Q1:

Software Quality Assurance (SQA) is the process of ensuring that software is developed according to defined standards and quality requirements. It focuses on preventing defects by improving the development process.

Quality is important because:-

- i) Ensures customer satisfaction.
- ii) Reduces software defects.
- iii) Lower development and maintenance costs.

iv) Improves reliability and security.

v) Makes software easier to maintain and enhances business reputation.

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QA

- Process Oriented.
- Focuses on preventing defects.
- Ensures the development process is followed correctly.
- Performed throughout the SDLC.
- Defining coding standards and development processes.

QC

- Product-oriented.
- Focuses on finding defects.
- Verifies the final product meets requirements.
- Performed after development or during testing.
- Checking whether the completed software contains bugs.

QC

- A broad quality verification process.
- Includes testing, code reviews, inspection and walkthrough.
- Ensures the product meets quality standards.
- Reviewing source code and performing software testing.

Testing

- A specific activity with QC.
- Involves executing software to detect defects.
- Verifies software behaves as expected.
- Running login test cases to verify connect functionality.

⇒ Testing is a subset of QC.

QA	Testing
→ Process-oriented. → prevents defects. → Covers the entire software development process. → Includes standards, audits, process improvement and training.	→ Execution oriented. → Detects bugs. → Focuses on checking the software by executing test cases. → Includes functional, integration, system and other test types.

⇒ Testing is a part of QC and QC is a part of QA.

QA: Prevent defects (improve process).

QC: Find defects (check the product).

Testing: Execute software to find bugs.

Q3:

Artificial Intelligence (AI) is transforming software testing by making it faster, smarter and more efficient. AI provides -

- i) Automated Test case Generation.
- ii) Self-Healing Test scripts.
- iii) Bug prediction.
- iv) Visual Testing.
- v) Faster Regression Testing.
- vi) Improved Defect Analysis.
- vii) Better Test coverage.

Advantages:

- 1) Faster testing and quicker releases.
- 2) Reduced manual effort.
- 3) Higher test accuracy.
- 4) Early defect detection.

5. Lower testing cost.

6. Improved software quality.

Limitations:

1. Requires quality training data.
2. Cannot fully replace human judgment, creativity and exploratory testing.

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QA Engineer: Ensures the software development process follows quality standards by creating QA plans, conducting audits and improving development process.

Software Test Engineer: Designs and executes test cases, reports defects, verifies bug fixes and ensures the software functions correctly.

AI Quality Engineer: Tests AI/ML

models, validates datasets, measures model performance, checks for bias and ensures AI systems produce accurate and reliable results.

Q5:

Career Road-map

